

Concept 2 | Quadratic Equations

English Student Edition • Focused formulas and guided practice

Formula opener

Roots

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Sum/product

$$\alpha + \beta = -\frac{b}{a}, \quad \alpha\beta = \frac{c}{a}$$

Discriminant

$$\Delta = b^2 - 4ac$$

Classified Practice

Quadratic Equations

Q001 Written

Solve $x^2 + 8 = 0$.

Q002 Written**Solve** $x^2 + 4 = 0$.

Classified Practice

Quadratic Equations

Q003 Written

Solve $x^2 + 49 = 0$.

Q004 Written**Solve** $9x^2 - 4 = 0$.

Classified Practice

Quadratic Equations

Q005 Written

Solve $x^2 - 12 = 0$.

Q006 Written**Solve** $4x^2 + 3 = 0$.

Classified Practice

Quadratic Equations

Q007 MCQ

Solve $x^2 + 8 = 0$ Select the correct answer:**Choose the correct option.**

A $x = \pm 2i\sqrt{2}$

B $x = 2i\sqrt{2}$

C $x = -2i\sqrt{2}$

D $x = +2i\sqrt{2}$

Q008 MCQ**Solve $x^2 + 4 = 0$ Select the correct answer:****Choose the correct option.**

A $x = \pm 2i$

B $x = 2i$

C $x = -2i$

D $x = +2i$

Classified Practice

Quadratic Equations

Q009 MCQ

Solve $x^2 + 49 = 0$ Select the correct answer:**Choose the correct option.**

A $x = -7i$

B $x = +7i$

C $x = \pm 7i$

D $x = 7i$

Q010 MCQ**Solve $9x^2 - 4 = 0$ Select the correct answer:****Choose the correct option.**

A $x = -\frac{2}{3}$

B $x = +\frac{2}{3}$

C $x = \pm\frac{2}{3}$

D $x = \frac{2}{3}$

Classified Practice

Quadratic Equations

Q011 MCQ

Solve $x^2 - 12 = 0$ Select the correct answer:**Choose the correct option.**

A $x = +2\sqrt{3}$

B $x = \pm 2\sqrt{3}$

C $x = 2\sqrt{3}$

D $x = -2\sqrt{3}$

Q012 MCQ

Solve $4x^2 + 3 = 0$ Select the correct answer:**Choose the correct option.**

A $x = +\frac{i\sqrt{3}}{2}$

B $x = \pm\frac{i\sqrt{3}}{2}$

C $x = \frac{i\sqrt{3}}{2}$

D $x = -\frac{i\sqrt{3}}{2}$

Classified Practice

Quadratic Equations

Q013 Written

Solve $(x - 3)^2 = -8$.

Q014 Written**Solve** $(x - 2)^2 = 5$.

Classified Practice

Quadratic Equations

Q015 **Written****Solve** $(x + 6)^2 = -6$.

Q016 Written**Solve** $(x + 4)^2 = -18$.

Classified Practice

Quadratic Equations

Q017 Written

Solve $(2x - 3)^2 = 7$.

Q018 MCQ**Solve $(x - 3)^2 = -8$ Select the correct answer:****Choose the correct option.**

A $x = 3 \pm 2i\sqrt{2}$

B $x = 3 \pm 2i\sqrt{2}$

C $x = 3 - 2i\sqrt{2}$

D $x = 3 + 2i\sqrt{2}$

Classified Practice

Quadratic Equations

Q019 MCQ

Solve $(x - 2)^2 = 5$ Select the correct answer:**Choose the correct option.**

A $x = 2 + \sqrt{5}$

B $x = 2 \pm \sqrt{5}$

C $x = 2\sqrt{5}$

D $x = 2 - \sqrt{5}$

Q020 MCQ

Solve $(x + 6)^2 = -6$ Select the correct answer:**Choose the correct option.**

A $x = -6 - i\sqrt{6}$

B $x = -6 + i\sqrt{6}$

C $x = -6 \pm i\sqrt{6}$

D $x = -6i\sqrt{6}$

Classified Practice

Quadratic Equations

Q021 MCQ

Solve $(x + 4)^2 = -18$ Select the correct answer:**Choose the correct option.**

A $x = -4 - 3i\sqrt{2}$

B $x = -4 + 3i\sqrt{2}$

C $x = -4 \pm 3i\sqrt{2}$

D $x = -43i\sqrt{2}$

Q022 MCQ

Solve $(2x - 3)^2 = 7$ Select the correct answer:**Choose the correct option.**

A $x = \frac{3\sqrt{7}}{2}$

B $x = \frac{3 - \sqrt{7}}{2}$

C $x = \frac{3 + \sqrt{7}}{2}$

D $x = \frac{3 \pm \sqrt{7}}{2}$

Classified Practice

Quadratic Equations

Q023 Written

Solve $2x^2 - 7x - 9 = 0$.

Q024 Written**Solve** $2x^2 + 5x - 3 = 0$.

Classified Practice

Quadratic Equations

Q025 Written

Solve $2x^2 + 7x = 0$.

Q026 Written**Solve** $2x^2 + 5x - 12 = 0$.

Classified Practice

Quadratic Equations

Q027 Written

Solve $x^2 - 8x + 15 = 0$.

Q028 MCQ

Solve $2x^2 - 7x - 9 = 0$ Select the correct answer:**Choose the correct option.**

A $x = -1, \frac{9}{2}$

B $-(x = -1, \frac{9}{2})$

C $x = -1, \frac{9}{2} + 1$

D $x = -1, \frac{9}{2} - 1$

Classified Practice

Quadratic Equations

Q029 MCQ

Solve $2x^2 + 5x - 3 = 0$ Select the correct answer:**Choose the correct option.**

A $-(x = \frac{1}{2}, -3)$

B $x = \frac{1}{2}, -3 + 1$

C $x = \frac{1}{2}, -3 - 1$

D $x = \frac{1}{2}, -3$

Q030 MCQ**Solve $2x^2 + 7x = 0$ Select the correct answer:****Choose the correct option.**

A $-(x = 0, -\frac{7}{2})$

B $x = 0, -\frac{7}{2} + 1$

C $x = 0, -\frac{7}{2} - 1$

D $x = 0, -\frac{7}{2}$

Classified Practice

Quadratic Equations

Q031 MCQ

Solve $2x^2 + 5x - 12 = 0$ Select the correct answer:**Choose the correct option.**

A $x = \frac{3}{2}, -4$

B $-(x = \frac{3}{2}, -4)$

C $x = \frac{3}{2}, -4 + 1$

D $x = \frac{3}{2}, -4 - 1$

Q032 MCQ**Solve $x^2 - 8x + 15 = 0$ Select the correct answer:****Choose the correct option.****A** $x = 3, 5$ **B** $x = 3, 6$ **C** $x = 3, 5 - 1$ **D** $x = 3, 5$ (sign error)

Classified Practice

Quadratic Equations

Q033 Written

Solve $3x^2 - 4x - 1 = 0$.

Q034 MCQ

Solve $3x^2 - 4x - 1 = 0$ Select the correct answer:**Choose the correct option.**

A $x = \frac{2 + \sqrt{7}}{3}$

B $x = \frac{2 \pm \sqrt{7}}{3}$

C $x = \frac{2\sqrt{7}}{3}$

D $x = \frac{2 - \sqrt{7}}{3}$

Classified Practice

Quadratic Equations

Q035 Written

Solve $2x^2 + 2\sqrt{6}x + 11 = 0$.

Q036 Written**Solve** $2x^2 + 2\sqrt{3}x + 5 = 0$.

Classified Practice

Quadratic Equations

Q037 Written

Solve $x^2 + x + 1 = 0$.

Q038 Written**Solve** $3x^2 - 5x + 3 = 0$.

Classified Practice

Quadratic Equations

Q039 Written

Solve $x^2 - 2\sqrt{3}x + 5 = 0$.

Q040 Written**Solve** $3x^2 + 2\sqrt{2}x + 1 = 0$.

Classified Practice

Quadratic Equations

Q041 Written

Solve $x^2 - x + 2 = 0$.

Q042 Written**Solve** $3x^2 - \sqrt{5}x + 1 = 0$.

Classified Practice

Quadratic Equations

Q043 Written

Solve $6x^2 - 12x + 15 = 0$.

Q044 MCQ

Solve $2x^2 + 2\sqrt{6}x + 11 = 0$ Select the correct answer:**Choose the correct option.**

A $x = \frac{-\sqrt{6} + 4i}{2}$

B $x = \frac{-\sqrt{6} \pm 4i}{2}$

C $x = \frac{-\sqrt{6}4i}{2}$

D $x = \frac{-\sqrt{6} - 4i}{2}$

Classified Practice

Quadratic Equations

Q045 MCQ

Solve $2x^2 + 2\sqrt{3}x + 5 = 0$ **Select the correct answer:****Choose the correct option.**

A $x = \frac{-\sqrt{3} - i\sqrt{7}}{2}$

B $x = \frac{-\sqrt{3} + i\sqrt{7}}{2}$

C $x = \frac{-\sqrt{3} \pm i\sqrt{7}}{2}$

D $x = \frac{-\sqrt{3}i\sqrt{7}}{2}$

Q046 MCQ

Solve $x^2 + x + 1 = 0$ Select the correct answer:**Choose the correct option.**

A $x = \frac{-1i\sqrt{3}}{2}$

B $x = \frac{-1 - i\sqrt{3}}{2}$

C $x = \frac{-1 + i\sqrt{3}}{2}$

D $x = \frac{-1 \pm i\sqrt{3}}{2}$

Classified Practice

Quadratic Equations

Q047 MCQ

Solve $3x^2 - 5x + 3 = 0$ **Select the correct answer:****Choose the correct option.**

A $x = \frac{5 + i\sqrt{11}}{6}$

B $x = \frac{5 \pm i\sqrt{11}}{6}$

C $x = \frac{5i\sqrt{11}}{6}$

D $x = \frac{5 - i\sqrt{11}}{6}$

Q048 MCQ**Solve $x^2 - 2\sqrt{3}x + 5 = 0$ Select the correct answer:****Choose the correct option.**

- A $x = \sqrt{3}i\sqrt{2}$
 - B $x = \sqrt{3} - i\sqrt{2}$
 - C $x = \sqrt{3} + i\sqrt{2}$
 - D $x = \sqrt{3} \pm i\sqrt{2}$
-
-
-

Classified Practice

Quadratic Equations

Q049 MCQ

Solve $3x^2 + 2\sqrt{2}x + 1 = 0$ **Select the correct answer:****Choose the correct option.**

A $x = \frac{-\sqrt{2} + i}{3}$

B $x = \frac{-\sqrt{2} \pm i}{3}$

C $x = \frac{-\sqrt{2}i}{3}$

D $x = \frac{-\sqrt{2} - i}{3}$

Q050 MCQ

Solve $x^2 - x + 2 = 0$ Select the correct answer:**Choose the correct option.**

A $x = \frac{1 - i\sqrt{7}}{2}$

B $x = \frac{1 + i\sqrt{7}}{2}$

C $x = \frac{1 \pm i\sqrt{7}}{2}$

D $x = \frac{1i\sqrt{7}}{2}$

Classified Practice

Quadratic Equations

Q051 MCQ

Solve $3x^2 - \sqrt{5}x + 1 = 0$ **Select the correct answer:****Choose the correct option.**

A $x = \frac{\sqrt{5} - i\sqrt{7}}{6}$

B $x = \frac{\sqrt{5} + i\sqrt{7}}{6}$

C $x = \frac{\sqrt{5} \pm i\sqrt{7}}{6}$

D $x = \frac{\sqrt{5}i\sqrt{7}}{6}$

Q052 MCQ

Solve $6x^2 - 12x + 15 = 0$ Select the correct answer:**Choose the correct option.**

A $x = 1 + \frac{i\sqrt{6}}{2}$

B $x = 1 \pm \frac{i\sqrt{6}}{2}$

C $x = 1 \frac{i\sqrt{6}}{2}$

D $x = 1 - \frac{i\sqrt{6}}{2}$

Classified Practice

Quadratic Equations

Q053 Written

Determine the nature of the solutions of $3x^2 + 4x + 7 = 0$.

Q054 Written**Find the range of m for which $x^2 - mx + m^2 - 3m - 9 = 0$ has real solutions.**

Classified Practice

Quadratic Equations

Q055 **Written**

For $x^2 + kx + 3 - k = 0$, determine the nature of the solutions as k varies.

Q056 Written**For $3x^2 - 5x + 4 = 0$, determine the nature of the solutions.**

Classified Practice

Quadratic Equations

Q057 **Written**

For $4x^2 - 7x - 2 = 0$, determine the nature of the solutions.

Q058 Written**For $25x^2 - 10x + 1 = 0$, determine the nature of the solutions.**

Classified Practice

Quadratic Equations

Q059 **Written****Determine the nature of the solutions of $x^2 - 5x + 3 = 0$.**

Q060 Written**Determine the nature of the solutions of $9x^2 - 12x + 4 = 0$.**

Classified Practice

Quadratic Equations

Q061 Written

Determine the nature of the solutions of $4x^2 - 8x + 5 = 0$.

Q062 Written**Determine the nature of the solutions of $4x^2 - 4x + 1 = 0$.**

Classified Practice

Quadratic Equations

Q063 Written

Determine the nature of the solutions of $x^2 - 3x + 5 = 0$.

Q064 Written**Determine the nature of the solutions of $x^2 - x - 2 = 0$.**

Classified Practice

Quadratic Equations

Q065 MCQ

Determine the nature of the solutions of $3x^2 + 4x + 7 = 0$ Select the correct answer:**Choose the correct option.**

- A Two distinct imaginary (non-real complex) solutions
- B Two repeated imaginary (non-real complex) solutions
- C Two distinct imaginary (non-imaginary complex) solutions
- D Two distinct real (non-real complex) solutions

Classified Practice

Quadratic Equations

Q066 MCQ

Find the range of m for which $x^2 - mx + m^2 - 3m - 9 = 0$ has real solutions Select the correct answer:

Choose the correct option.

A $-2 \geq m \geq 6$

B $+2 \leq m \leq 6$

C $-2 \leq m \leq 7$

D $-2 \leq m \leq 6$

Classified Practice

Quadratic Equations

Q067 MCQ

For $x^2 + kx + 3 - k = 0$, determine the nature of the solutions as k varies. Select the correct answer:

Choose the correct option.

- A $k < -6$ or $k > 2$: distinct imaginary; $k = -6, 2$: repeated imaginary; $-6 < k < 2$: distinct imaginary
- B $k < -6$ or $k > 2$: distinct real; $k = -6, 2$: repeated real; $-6 < k < 2$: distinct real
- C $k < -6$ or $k > 2$: distinct real; $k = -6, 2$: repeated real; $-6 < k < 2$: distinct imaginary
- D $k < -6$ or $k > 2$: repeated real; $k = -6, 2$: repeated real; $-6 < k < 2$: repeated imaginary

Classified Practice

Quadratic Equations

Q068 MCQ

For $3x^2 - 5x + 4 = 0$, determine the nature of the solutions Select the correct answer:

Choose the correct option.

- A Two distinct imaginary (non-imaginary complex) solutions
- B Two distinct real (non-real complex) solutions
- C Two distinct imaginary (non-real complex) solutions
- D Two repeated imaginary (non-real complex) solutions

Classified Practice

Quadratic Equations

Q069 MCQ

For $4x^2 - 7x - 2 = 0$, determine the nature of the solutions Select the correct answer:

Choose the correct option.

- A Two distinct real solutions
- B Two repeated real solutions
- C Two distinct imaginary solutions
- D Two distinct real solutions;(sign error)

Classified Practice

Quadratic Equations

Q070 MCQ

For $25x^2 - 10x + 1 = 0$, determine the nature of the solutions Select the correct answer:

Choose the correct option.

A One repeated real solution

B One repeated real solution;(sign error)

C 0

D 1

Classified Practice

Quadratic Equations

Q071 MCQ

Determine the nature of the solutions of $x^2 - 5x + 3 = 0$ Select the correct answer:**Choose the correct option.**

- A Two distinct imaginary solutions
- B Two distinct real solutions;(sign error)
- C Two distinct real solutions
- D Two repeated real solutions

Classified Practice

Quadratic Equations

Q072 MCQ

Determine the nature of the solutions of $9x^2 - 12x + 4 = 0$ Select the correct answer:**Choose the correct option.**

A 0

B 1

C One repeated real solution

D One repeated real solution;(sign error)

Classified Practice

Quadratic Equations

Q073 MCQ

Determine the nature of the solutions of $4x^2 - 8x + 5 = 0$ Select the correct answer:**Choose the correct option.**

- A Two distinct real (non-real complex) solutions
- B Two distinct imaginary (non-real complex) solutions
- C Two repeated imaginary (non-real complex) solutions
- D Two distinct imaginary (non-imaginary complex) solutions

Classified Practice

Quadratic Equations

Q074 MCQ

Determine the nature of the solutions of $4x^2 - 4x + 1 = 0$ Select the correct answer:**Choose the correct option.**

A One repeated real solution

B One repeated real solution;(sign error)

C 0

D 1

Classified Practice

Quadratic Equations

Q075 MCQ

Determine the nature of the solutions of $x^2 - 3x + 5 = 0$ Select the correct answer:**Choose the correct option.**

- A Two distinct imaginary (non-real complex) solutions
- B Two repeated imaginary (non-real complex) solutions
- C Two distinct imaginary (non-imaginary complex) solutions
- D Two distinct real (non-real complex) solutions

Classified Practice

Quadratic Equations

Q076 MCQ

Determine the nature of the solutions of $x^2 - x - 2 = 0$ Select the correct answer:**Choose the correct option.**

- A Two distinct real solutions
- B Two repeated real solutions
- C Two distinct imaginary solutions
- D Two distinct real solutions;(sign error)

Classified Practice

Quadratic Equations

Q077 Written

Find the range of m for which $x^2 - 2mx + m + 2 = 0$ has two distinct imaginary solutions.

Q078 Written

For $x^2 - ax + 2a - 3 = 0$, determine the nature of the solutions as a varies.

Classified Practice

Quadratic Equations

Q079 **Written**

Find the values of a for which $x^2 - (a + 1)x + 4 = 0$ has a repeated solution.

Q080 Written

Find the range of m for which $x^2 + (m + 6)x - 2m = 0$ has two distinct imaginary solutions.

Classified Practice

Quadratic Equations

Q081 Written

Find the range of m for which $x^2 - 2mx + 2m^2 - 3m = 0$ has real solutions.

Q082 MCQ

Find the range of m for which $x^2 - 2mx + m + 2 = 0$ has two distinct imaginary solutions Select the correct answer:

Choose the correct option.

A $+1 < m < 2$

B $-1 < m < 3$

C $-1 < m < 2$

D $-1 > m > 2$

Classified Practice

Quadratic Equations

Q083 MCQ

For $x^2 - ax + 2a - 3 = 0$, determine the nature of the solutions as a varies. Select the correct answer:

Choose the correct option.

A $a < 2$ or $a > 6$: distinct imaginary; $a = 2, 6$: repeated imaginary; $2 < a < 6$: distinct imaginary

B $a < 2$ or $a > 6$: distinct real; $a = 2, 6$: repeated real; $2 < a < 6$: distinct real

C $a < 2$ or $a > 6$: distinct real; $a = 2, 6$: repeated real; $2 < a < 6$: distinct imaginary

D $a < 2$ or $a > 6$: repeated real; $a = 2, 6$: repeated real; $2 < a < 6$: repeated imaginary

Classified Practice

Quadratic Equations

Q084 MCQ

Find the values of a for which $x^2 - (a + 1)x + 4 = 0$ has a repeated solution Select the correct answer:

Choose the correct option.

A $a = 3$ or $a = -5$

B $a = 3$ or $a = -4$

C $a = 3$ or $a = -5 - 1$

D $a = 3$ or $a = -5$ (sign error)

Q085 MCQ

Find the range of m for which $x^2 + (m + 6)x - 2m = 0$ has two distinct imaginary solutions Select the correct answer:

Choose the correct option.

A $-18 > m > -2$

B $+18 < m < -2$

C $-18 < m < -1$

D $-18 < m < -2$

Classified Practice

Quadratic Equations

Q086 MCQ

Find the range of m for which $x^2 - 2mx + 2m^2 - 3m = 0$ has real solutions Select the correct answer:

Choose the correct option.

A $0 \leq m \leq 4$

B $0 \leq m \leq 3 - 1$

C $0 \leq m \leq 3$

D $0 \geq m \geq 3$

Q087 Written**Find the sum and product of the two roots of $3x^2 - 6x - 5 = 0$.**

Classified Practice

Quadratic Equations

Q088 Written

Find the sum and product of the roots of $4x^2 + 2x - 7 = 0$.

Q089 Written**Find the sum and product of the roots of $x^2 + x + 2 = 0$.**

Classified Practice

Quadratic Equations

Q090 Written

Find the sum and product of the roots of $3x^2 - 7x - 6 = 0$.

Q091 MCQ

Find the sum and product of the two roots of $3x^2 - 6x - 5 = 0$ Select the correct answer:

Choose the correct option.

- A $\alpha + \beta = 2, \quad \alpha\beta = -\frac{5}{3} + 1$
- B $\alpha + \beta = 2, \quad \alpha\beta = -\frac{5}{3} - 1$
- C $\alpha + \beta = 2, \quad \alpha\beta = -\frac{5}{3}$
- D $-(\alpha + \beta = 2, \quad \alpha\beta = -\frac{5}{3})$

Classified Practice

Quadratic Equations

Q092 MCQ

Find the sum and product of the roots of $4x^2 + 2x - 7 = 0$ Select the correct answer:**Choose the correct option.**

- A $\alpha + \beta = -\frac{1}{2}, \quad \alpha\beta = -\frac{7}{4}$
- B $-(\alpha + \beta = -\frac{1}{2}, \quad \alpha\beta = -\frac{7}{4})$
- C $\alpha + \beta = -\frac{1}{2}, \quad \alpha\beta = -\frac{7}{4} + 1$
- D $\alpha + \beta = -\frac{1}{2}, \quad \alpha\beta = -\frac{7}{4} - 1$

Q093 MCQ

Find the sum and product of the roots of $x^2 + x + 2 = 0$ Select the correct answer:

Choose the correct option.

A $S = -1$, $P = 2$ (sign error)

B $S = -1$, $P = 2$

C $S = -1$, $P = 3$

D $S = -1$, $P = 2 - 1$

Classified Practice

Quadratic Equations

Q094 MCQ

Find the sum and product of the roots of $3x^2 - 7x - 6 = 0$ Select the correct answer:**Choose the correct option.**

A $S = \frac{7}{3}, P = -2$

B $-(S = \frac{7}{3}, P = -2)$

C $S = \frac{7}{3}, P = -2 + 1$

D $S = \frac{7}{3}, P = -2 - 1$

Q095 Written

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $\alpha^2 + \beta^2$.

Classified Practice

Quadratic Equations

Q096 Written

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $\alpha^3 + \beta^3$.

Q097 Written

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $\alpha^2 + \beta^2$.

Classified Practice

Quadratic Equations

Q098 Written

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $\alpha^3 + \beta^3$.

Q099 Written**If α, β are roots of $x^2 - 2x + 3 = 0$, find $\alpha^2 + \beta^2$.**

Classified Practice

Quadratic Equations

Q100 Written

If α, β are roots of $x^2 - 2x + 3 = 0$, find $\alpha^3 + \beta^3$.

Q101 Written

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $\alpha^2 + \beta^2$.

Classified Practice

Quadratic Equations

Q102 Written

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $\alpha^3 + \beta^3$.

Q103 MCQ

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $\alpha^2 + \beta^2$ Select the correct answer:

Choose the correct option.

A 12

B 13

C $12 - 1$

D 12 (sign error)

Classified Practice

Quadratic Equations

Q104 MCQ

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $\alpha^3 + \beta^3$ Select the correct answer:

Choose the correct option.

- A $\frac{81}{2} - 1$
B $\frac{81}{2}$
C $-(\frac{81}{2})$
D $\frac{81}{2} + 1$

Q105 MCQ

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $\alpha^2 + \beta^2$ Select the correct answer:

Choose the correct option.

A 1

B 2

C $1 - 1$

D 1 (sign error)

Classified Practice

Quadratic Equations

Q106 MCQ

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $\alpha^3 + \beta^3$ Select the correct answer:

Choose the correct option.

A -1 (sign error)

B -1

C 0

D $-1 - 1$

Q107 MCQ

If α, β are roots of $x^2 - 2x + 3 = 0$, find $\alpha^2 + \beta^2$ Select the correct answer:

Choose the correct option.

A -1

B -2 - 1

C -2 (sign error)

D -2

Classified Practice

Quadratic Equations

Q108 MCQ

If α, β are roots of $x^2 - 2x + 3 = 0$, find $\alpha^3 + \beta^3$ Select the correct answer:

Choose the correct option.

A -10 (sign error)

B -10

C -9

D $-10 - 1$

Q109 MCQ

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $\alpha^2 + \beta^2$ Select the correct answer:

Choose the correct option.

A $\frac{14}{3} - 1$

B $\frac{14}{3}$

C $-(\frac{14}{3})$

D $\frac{14}{3} + 1$

Classified Practice

Quadratic Equations

Q110 MCQ

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $\alpha^3 + \beta^3$ Select the correct answer:

Choose the correct option.

A -9

B $-10 - 1$

C -10 (sign error)

D -10

Q111 Written

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $\frac{2}{\alpha} + \frac{2}{\beta}$.

Classified Practice

Quadratic Equations

Q112 Written

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $\frac{\beta}{\alpha} + \frac{\alpha}{\beta}$.

Q113 Written

Find the sum and product of the roots of $5x^2 - 3 = 0$.

Classified Practice

Quadratic Equations

Q114 **Written**

If α, β are roots of $x^2 - 2x + 3 = 0$, find $\frac{1}{\alpha} + \frac{1}{\beta}$.

Q115 Written

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $\frac{1}{\alpha} + \frac{1}{\beta}$.

Classified Practice

Quadratic Equations

Q116 MCQ

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $\frac{2}{\alpha} + \frac{2}{\beta}$ Select the correct answer:

Choose the correct option.

A $-4 - 1$

B -4 (sign error)

C -4

D -3

Q117 MCQ

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $\frac{\beta}{\alpha} + \frac{\alpha}{\beta}$ Select the correct answer:

Choose the correct option.

A $-(\frac{2}{3})$

B $\frac{2}{3} + 1$

C $\frac{2}{3} - 1$

D $\frac{2}{3}$

Classified Practice

Quadratic Equations

Q118 MCQ

Find the sum and product of the roots of $5x^2 - 3 = 0$ Select the correct answer:**Choose the correct option.**

A $-(S = 0, \quad P = -\frac{3}{5})$

B $S = 0, \quad P = -\frac{3}{5} + 1$

C $S = 0, \quad P = -\frac{3}{5} - 1$

D $S = 0, \quad P = -\frac{3}{5}$

Q119 MCQ

If α, β are roots of $x^2 - 2x + 3 = 0$, find $\frac{1}{\alpha} + \frac{1}{\beta}$ Select the correct answer:

Choose the correct option.

A $-\left(\frac{2}{3}\right)$

B $\frac{2}{3} + 1$

C $\frac{2}{3} - 1$

D $\frac{2}{3}$

Classified Practice

Quadratic Equations

Q120 MCQ

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $\frac{1}{\alpha} + \frac{1}{\beta}$ Select the correct answer:

Choose the correct option.

A 7

B $6 - 1$

C 6 (sign error)

D 6

Q121 Written

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $(\alpha + 1)(\beta + 1)$.

Classified Practice

Quadratic Equations

Q122 Written

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $(\alpha - 1)(\beta - 1)$.

Q123 Written

If α, β are roots of $x^2 - 2x + 3 = 0$, find $(\alpha + 2)(\beta + 2)$.

Classified Practice

Quadratic Equations

Q124 **Written**

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $(\alpha - 2)(\beta - 2)$.

Q125 MCQ

If α, β are roots of $2x^2 - 6x - 3 = 0$, find $(\alpha + 1)(\beta + 1)$ Select the correct answer:

Choose the correct option.

- A $\frac{5}{2}$
- B $-(\frac{5}{2})$
- C $\frac{5}{2} + 1$
- D $\frac{5}{2} - 1$

Classified Practice

Quadratic Equations

Q126 MCQ

If α, β are roots of $2x^2 - 4x + 3 = 0$, find $(\alpha - 1)(\beta - 1)$ Select the correct answer:

Choose the correct option.

A $\frac{1}{2} + 1$

B $\frac{1}{2} - 1$

C $\frac{1}{2}$

D $-(\frac{1}{2})$

Q127 MCQ

If α, β are roots of $x^2 - 2x + 3 = 0$, find $(\alpha + 2)(\beta + 2)$ Select the correct answer:

Choose the correct option.

A 12

B $11 - 1$

C 11 (sign error)

D 11

Classified Practice

Quadratic Equations

Q128 MCQ

If α, β are roots of $3x^2 + 6x - 1 = 0$, find $(\alpha - 2)(\beta - 2)$ Select the correct answer:

Choose the correct option.

A $-(\frac{23}{3})$

B $\frac{23}{3} + 1$

C $\frac{23}{3} - 1$

D $\frac{23}{3}$

Q129 Written

If one root of $x^2 - 8x + k = 0$ is three times the other, find k and the roots.

Classified Practice

Quadratic Equations

Q130 **Written**

If one root of $x^2 + 6x + m = 0$ is twice the other, find m and the roots.

Q131 Written

If one root of $8x^2 - mx + 1 = 0$ is twice the other, find m and the roots.

Classified Practice

Quadratic Equations

Q132 MCQ

If one root of $x^2 - 8x + k = 0$ is three times the other, find k and the roots Select the correct answer:

Choose the correct option.

A $k = 6, \quad x = 2, 6$

B $k = 6, \quad x = 2, 7$

C $k = 6, \quad x = 2, 6 - 1$

D $k = 6, \quad x = 2, 6$ (sign error)

Q133 MCQ

If one root of $x^2 + 6x + m = 0$ is twice the other, find m and the roots Select the correct answer:

Choose the correct option.

- A $m = 8, \quad x = -2, -4$
- B $m = 8, \quad x = -2, -3$
- C $m = 8, \quad x = -2, -4 - 1$
- D $m = 8, \quad x = -2, -4$ (sign error)

Classified Practice

Quadratic Equations

Q134 MCQ

If one root of $8x^2 - mx + 1 = 0$ is twice the other, find m and the roots. Select the correct answer:

Choose the correct option.

- A $(m, x_1, x_2) = (6, \frac{1}{4}, \frac{1}{2})$ or $(-6, -\frac{1}{4}, -\frac{1}{3})$
- B $(m, x_1, x_2) = (6, \frac{1}{4}, \frac{1}{2})$ or $(-6, -\frac{1}{4}, -\frac{1}{2}) - 1$
- C $(m, x_1, x_2) = (6, \frac{1}{4}, \frac{1}{2})$ or $(-6, -\frac{1}{4}, -\frac{1}{2})$ (sign error)
- D $(m, x_1, x_2) = (6, \frac{1}{4}, \frac{1}{2})$ or $(-6, -\frac{1}{4}, -\frac{1}{2})$

Classified Practice

Quadratic Equations

Q135 Written

If the difference between the two roots of $x^2 - (m - 1)x - 6 = 0$ is 5, find m and the roots.

Q136 Written

If the difference between the roots of $x^2 - (m + 1)x + 2 = 0$ is 1, find m and the roots.

Classified Practice

Quadratic Equations

Q137 Written

If the difference between the roots of $x^2 + 6x + m = 0$ is 4, find m and the roots.

Q138 Written

If the difference between the roots of $4x^2 + mx + 3 = 0$ is 1, find m and the roots.

Classified Practice

Quadratic Equations

Q139 MCQ

If the difference between the two roots of $x^2 - (m - 1)x - 6 = 0$ is 5, find m and the roots. Select the correct answer:

Choose the correct option.

- A $(m, x_1, x_2) = (2, -2, 3)$ or $(0, -3, 3)$
- B $(m, x_1, x_2) = (2, -2, 3)$ or $(0, -3, 2) - 1$
- C $(m, x_1, x_2) = (2, -2, 3)$ or $(0, -3, 2)$ (sign error)
- D $(m, x_1, x_2) = (2, -2, 3)$ or $(0, -3, 2)$

Classified Practice

Quadratic Equations

Q140 MCQ

If the difference between the roots of $x^2 - (m + 1)x + 2 = 0$ is 1, find m and the roots Select the correct answer:

Choose the correct option.

- A $(m, x_1, x_2) = (2, 1, 2)$ or $(-4, -2, -1)$
- B $(m, x_1, x_2) = (2, 1, 2)$ or $(-4, -2, 0)$
- C $(m, x_1, x_2) = (2, 1, 2)$ or $(-4, -2, -1) - 1$
- D $(m, x_1, x_2) = (2, 1, 2)$ or $(-4, -2, -1)$ (sign error)

Classified Practice

Quadratic Equations

Q141 MCQ

If the difference between the roots of $x^2 + 6x + m = 0$ is 4, find m and the roots Select the correct answer:

Choose the correct option.

A $m = 5$, $x = -5, -1$ (sign error)

B $m = 5$, $x = -5, -1$

C $m = 5$, $x = -5, 0$

D $m = 5$, $x = -5, -1 - 1$

Classified Practice

Quadratic Equations

Q142 MCQ

If the difference between the roots of $4x^2 + mx + 3 = 0$ is 1, find m and the roots Select the correct answer:

Choose the correct option.

A $(m, x_1, x_2) = (-8, \frac{1}{2}, \frac{3}{2})$ or $(8, -\frac{3}{2}, -\frac{1}{2})$ (sign error)

B $(m, x_1, x_2) = (-8, \frac{1}{2}, \frac{3}{2})$ or $(8, -\frac{3}{2}, -\frac{1}{2})$

C $(m, x_1, x_2) = (-8, \frac{1}{2}, \frac{3}{2})$ or $(8, -\frac{3}{2}, -\frac{1}{2})$

D $(m, x_1, x_2) = (-8, \frac{1}{2}, \frac{3}{2})$ or $(8, -\frac{3}{2}, -\frac{1}{2}) - 1$

Classified Practice

Quadratic Equations

Q143 Written

If the ratio of the roots of $x^2 - 2x + m = 0$ is $2 : 3$, find m and the roots.

Q144 MCQ

If the ratio of the roots of $x^2 - 2x + m = 0$ is $2 : 3$, find m and the roots Select the correct answer:

Choose the correct option.

- A $m = \frac{24}{25}, \quad x = \frac{4}{5}, \frac{6}{5}$
B $-(m = \frac{24}{25}, \quad x = \frac{4}{5}, \frac{6}{5})$
C $m = \frac{24}{25}, \quad x = \frac{4}{5}, \frac{6}{5} + 1$
D $m = \frac{24}{25}, \quad x = \frac{4}{5}, \frac{6}{5} - 1$

Classified Practice

Quadratic Equations

Q145 **Written**

If one root of $x^2 - 6x + k = 0$ is the square of the other, find k and the roots.

Classified Practice

Quadratic Equations

Q146 MCQ

If one root of $x^2 - 6x + k = 0$ is the square of the other, find k and the roots. Select the correct answer:

Choose the correct option.

A $(k, x_1, x_2) = (8, 2, 4)$ or $(-27, -3, 9)$ (sign error)

B $(k, x_1, x_2) = (8, 2, 4)$ or $(-27, -3, 9)$

C $(k, x_1, x_2) = (8, 2, 4)$ or $(-27, -3, 10)$

D $(k, x_1, x_2) = (8, 2, 4)$ or $(-27, -3, 9) - 1$

Classified Practice

Quadratic Equations

Q147 **Written****Factorise $2x^2 - 3x + 2$ over the complex numbers.**

Q148 Written**Factorise $x^2 + 2x + 5$ over the complex numbers.**

Classified Practice

Quadratic Equations

Q149 **Written****Factorise $2x^2 - 6x + 1$ over the complex numbers.**

Q150 Written**Factorise $x^2 + 4x + 5$ over the complex numbers.**

Classified Practice

Quadratic Equations

Q151 **Written****Factorise $x^2 + 4x + 2$ over the complex numbers.**

Q152 Written**Factorise $x^2 - 4x + 7$ over the complex numbers.**

Classified Practice

Quadratic Equations

Q153 **Written****Factorise $2x^2 - x + 1$ over the complex numbers.**

Q154 Written**Factorise $3x^2 - 5x + 3$ over the complex numbers.**

Classified Practice

Quadratic Equations

Q155 **Written****Factorise $2x^2 - 2x - 1$ over the complex numbers.**

Classified Practice

Quadratic Equations

Q156 MCQ

Factorise $2x^2 - 3x + 2$ over the complex numbers Select the correct answer:**Choose the correct option.**

A $2 \left(x - \frac{3 - i\sqrt{7}}{4} \right) \left(x - \frac{3 - i\sqrt{7}}{4} \right)$

B $-(2 \left(x - \frac{3 + i\sqrt{7}}{4} \right) \left(x - \frac{3 - i\sqrt{7}}{4} \right))$

C $2 \left(x - \frac{3 + i\sqrt{7}}{4} \right) \left(x - \frac{3 - i\sqrt{7}}{4} \right)$

D $2 \left(x - \frac{3 + \sqrt{7}}{4} \right) \left(x - \frac{3 - i\sqrt{7}}{4} \right)$

Classified Practice

Quadratic Equations

Q157 MCQ

Factorise $x^2 + 2x + 5$ over the complex numbers Select the correct answer:**Choose the correct option.**

A $(x + 1 - 2i)(x + 1 + 2i)$

B $(x + 1 - 2)(x + 1 + 2i)$

C $(x + 1 - 2 - i)(x + 1 + 2i)$

D $(x + 1 - 2i)(x + 13i)$

Classified Practice

Quadratic Equations

Q158 MCQ

Factorise $2x^2 - 6x + 1$ over the complex numbers Select the correct answer:**Choose the correct option.**

A $-(2\left(x - \frac{3 + \sqrt{7}}{2}\right)\left(x - \frac{3 - \sqrt{7}}{2}\right))$

B $2\left(x - \frac{3 + \sqrt{7}}{2}\right)\left(x - \frac{3 - \sqrt{7}}{2}\right) + 1$

C $2\left(x - \frac{3 + \sqrt{7}}{2}\right)\left(x - \frac{3 - \sqrt{7}}{2}\right) - 1$

D $2\left(x - \frac{3 + \sqrt{7}}{2}\right)\left(x - \frac{3 - \sqrt{7}}{2}\right)$

Classified Practice

Quadratic Equations

Q159 MCQ

Factorise $x^2 + 4x + 5$ over the complex numbers Select the correct answer:**Choose the correct option.**

A $(x + 2 - i)(x + 2 + i)$

B $(x + 2 + i)(x + 2 + i)$

C $(x + 2 - i)(x + 2 + i)$

D $(x + 2 - i)(x + 2 + i)$

Q160 MCQ

Factorise $x^2 + 4x + 2$ over the complex numbers Select the correct answer:

Choose the correct option.

- A $(x + 2 - \sqrt{2})(x + 2 + \sqrt{2})$ (sign error)
 - B $(x + 2 - \sqrt{2})(x + 2 + \sqrt{2})$
 - C $(x + 2 - \sqrt{2})(x + 2 + \sqrt{3})$
 - D $(x + 2 - \sqrt{2})(x + 2 + \sqrt{2}) - 1$
-
-
-

Classified Practice

Quadratic Equations

Q161 MCQ

Factorise $x^2 - 4x + 7$ over the complex numbers Select the correct answer:**Choose the correct option.**

A $(x - 2 - i\sqrt{3})(x - 2 + i\sqrt{3})$

B $(x - 2 - \sqrt{3})(x - 2 + i\sqrt{3})$

C $(x - 2 + i\sqrt{3})(x - 2 + i\sqrt{3})$

D $(x - 2 - i\sqrt{3})(x - 2 + i\sqrt{4})$

Classified Practice

Quadratic Equations

Q162 MCQ

Factorise $2x^2 - x + 1$ over the complex numbers Select the correct answer:**Choose the correct option.**

- A $2\left(x - \frac{1 - i\sqrt{7}}{4}\right)\left(x - \frac{1 - i\sqrt{7}}{4}\right)$
- B $-(2\left(x - \frac{1 + i\sqrt{7}}{4}\right)\left(x - \frac{1 - i\sqrt{7}}{4}\right))$
- C $2\left(x - \frac{1 + i\sqrt{7}}{4}\right)\left(x - \frac{1 - i\sqrt{7}}{4}\right)$
- D $2\left(x - \frac{1 + \sqrt{7}}{4}\right)\left(x - \frac{1 - i\sqrt{7}}{4}\right)$
-
-
-

Classified Practice

Quadratic Equations

Q163 MCQ

Factorise $3x^2 - 5x + 3$ over the complex numbers Select the correct answer:**Choose the correct option.**

A $3\left(x - \frac{5 + i\sqrt{11}}{6}\right)\left(x - \frac{5 - i\sqrt{11}}{6}\right)$

B $3\left(x - \frac{5 + \sqrt{11}}{6}\right)\left(x - \frac{5 - i\sqrt{11}}{6}\right)$

C $3\left(x - \frac{5 - i\sqrt{11}}{6}\right)\left(x - \frac{5 - i\sqrt{11}}{6}\right)$

D $-(3\left(x - \frac{5 + i\sqrt{11}}{6}\right)\left(x - \frac{5 - i\sqrt{11}}{6}\right))$

Classified Practice

Quadratic Equations

Q164 MCQ

Factorise $2x^2 - 2x - 1$ over the complex numbers Select the correct answer:**Choose the correct option.**

A $2\left(x - \frac{1 + \sqrt{3}}{2}\right)\left(x - \frac{1 - \sqrt{3}}{2}\right) - 1$

B $2\left(x - \frac{1 + \sqrt{3}}{2}\right)\left(x - \frac{1 - \sqrt{3}}{2}\right)$

C $-(2\left(x - \frac{1 + \sqrt{3}}{2}\right)\left(x - \frac{1 - \sqrt{3}}{2}\right))$

D $2\left(x - \frac{1 + \sqrt{3}}{2}\right)\left(x - \frac{1 - \sqrt{3}}{2}\right) + 1$

Classified Practice

Quadratic Equations

Q165 **Written****Factorise $x^4 + 2x^2 - 3$ over the complex numbers.**

Q166 Written**Factorise $x^4 - x^2 - 6$ over the complex numbers.**

Classified Practice

Quadratic Equations

Q167 **Written****Factorise $x^4 - 9$ over the complex numbers.**

Q168 Written**Factorise $x^4 - 3x^2 + 2$ over the complex numbers.**

Classified Practice

Quadratic Equations

Q169 MCQ

Factorise $x^4 + 2x^2 - 3$ over the complex numbers Select the correct answer:**Choose the correct option.**

A $(x - 1)(x + 1)(x - i\sqrt{3})(x + i\sqrt{4})$

B $(x - 1)(x + 1)(x - i\sqrt{3})(x + i\sqrt{3})$

C $(x - 1)(x + 1)(x - \sqrt{3})(x + i\sqrt{3})$

D $(x - 1)(x + 1)(x + i\sqrt{3})(x + i\sqrt{3})$

Q170 MCQ

Factorise $x^4 - x^2 - 6$ over the complex numbers Select the correct answer:

Choose the correct option.

A $(x - \sqrt{3})(x + \sqrt{3})(x - i\sqrt{2})(x + i\sqrt{3})$

B $(x - \sqrt{3})(x + \sqrt{3})(x - i\sqrt{2})(x + i\sqrt{2})$

C $(x - \sqrt{3})(x + \sqrt{3})(x - \sqrt{2})(x + i\sqrt{2})$

D $(x - \sqrt{3})(x + \sqrt{3})(x + i\sqrt{2})(x + i\sqrt{2})$

Classified Practice

Quadratic Equations

Q171 MCQ

Factorise $x^4 - 9$ over the complex numbers Select the correct answer:**Choose the correct option.**

- A $(x - \sqrt{3})(x + \sqrt{3})(x - \sqrt{3})(x + i\sqrt{3})$
B $(x - \sqrt{3})(x + \sqrt{3})(x + i\sqrt{3})(x + i\sqrt{3})$
C $(x - \sqrt{3})(x + \sqrt{3})(x - i\sqrt{3})(x + i\sqrt{4})$
D $(x - \sqrt{3})(x + \sqrt{3})(x - i\sqrt{3})(x + i\sqrt{3})$
-
-
-

Q172 MCQ

Factorise $x^4 - 3x^2 + 2$ over the complex numbers Select the correct answer:

Choose the correct option.

- A $(x - 1)(x + 1)(x - \sqrt{2})(x + \sqrt{2}) - 1$
 - B $(x - 1)(x + 1)(x - \sqrt{2})(x + \sqrt{2})$ (sign error)
 - C $(x - 1)(x + 1)(x - \sqrt{2})(x + \sqrt{2})$
 - D $(x - 1)(x + 1)(x - \sqrt{2})(x + \sqrt{3})$
-
-
-

Classified Practice

Quadratic Equations

Q173 Written

Create a quadratic equation with integer coefficients having $\frac{1+i}{2}$ and $\frac{1-i}{2}$ as roots.

Q174 Written**Find two numbers whose sum is -1 and product is 2 .**

Classified Practice

Quadratic Equations

Q175 Written

Create a quadratic equation with roots $3 + 2i$ and $3 - 2i$.

Q176 Written**Find two numbers whose sum and product are both 1.**

Classified Practice

Quadratic Equations

Q177 **Written****Create a quadratic equation with roots -1 and 2 .**

Q178 Written

Create a quadratic equation with roots $1 + 4i$ and $1 - 4i$.

Classified Practice

Quadratic Equations

Q179 **Written**

Create an integer-coefficient quadratic with roots $\frac{5+\sqrt{7}}{3}$ and $\frac{5-\sqrt{7}}{3}$.

Q180 Written

Find two numbers whose sum is 2 and product is 5.

Classified Practice

Quadratic Equations

Q181 **Written**

Find two numbers whose sum is 4 and product is 2.

Q182 MCQ

Create a quadratic equation with integer coefficients having $\frac{1+i}{2}$ and $\frac{1-i}{2}$ as roots Select the correct answer:

Choose the correct option.

A $2x^2 - 2x + 1 = 0$ (sign error)

B $2x^2 - 2x + 1 = 0$

C $2x^2 - 2x + 1 = 1$

D $2x^2 - 2x + 1 = 0 - 1$

Classified Practice

Quadratic Equations

Q183 MCQ

Find two numbers whose sum is -1 and product is 2 Select the correct answer:**Choose the correct option.**

A $-\left(\frac{-1 + i\sqrt{7}}{2}, \frac{-1 - i\sqrt{7}}{2}\right)$

B $\frac{-1 + i\sqrt{7}}{2}, \frac{-1 - i\sqrt{7}}{2}$

C $\frac{-1 + \sqrt{7}}{2}, \frac{-1 - i\sqrt{7}}{2}$

D $\frac{-1 - i\sqrt{7}}{2}, \frac{-1 - i\sqrt{7}}{2}$

Q184 MCQ

Create a quadratic equation with roots $3 + 2i$ and $3 - 2i$ Select the correct answer:

Choose the correct option.

A $x^2 - 6x + 13 = 0$ (sign error)

B $x^2 - 6x + 13 = 0$

C $x^2 - 6x + 13 = 1$

D $x^2 - 6x + 13 = 0 - 1$

Classified Practice

Quadratic Equations

Q185 MCQ

Find two numbers whose sum and product are both 1 Select the correct answer:**Choose the correct option.**

- A $\frac{1+i\sqrt{3}}{2}, \frac{1-i\sqrt{3}}{2}$
- B $\frac{1+\sqrt{3}}{2}, \frac{1-i\sqrt{3}}{2}$
- C $\frac{1-i\sqrt{3}}{2}, \frac{1-i\sqrt{3}}{2}$
- D $-(\frac{1+i\sqrt{3}}{2}, \frac{1-i\sqrt{3}}{2})$

Q186 MCQ

Create a quadratic equation with roots -1 and 2 Select the correct answer:

Choose the correct option.

A $x^2 - x - 2 = 0$

B $x^2 - x - 2 = 1$

C $x^2 - x - 2 = 0 - 1$

D $x^2 - x - 2 = 0$ (sign error)

Classified Practice

Quadratic Equations

Q187 MCQ

Create a quadratic equation with roots $1 + 4i$ and $1 - 4i$ Select the correct answer:**Choose the correct option.**

A $x^2 - 2x + 17 = 1$

B $x^2 - 2x + 17 = 0 - 1$

C $x^2 - 2x + 17 = 0$ (sign error)

D $x^2 - 2x + 17 = 0$

Q188 MCQ

Create an integer-coefficient quadratic with roots $\frac{5+\sqrt{7}}{3}$ and $\frac{5-\sqrt{7}}{3}$. Select the correct answer:

Choose the correct option.

A $3x^2 - 10x + 6 = 0$

B $3x^2 - 10x + 6 = 1$

C $3x^2 - 10x + 6 = 0 - 1$

D $3x^2 - 10x + 6 = 0$ (sign error)

Classified Practice

Quadratic Equations

Q189 MCQ

Find two numbers whose sum is 2 and product is 5 Select the correct answer:**Choose the correct option.**A $1 + 2i$, $1 - 1i$ B $1 + 2i$, $1 - 2i$ C $1 + 2$, $1 - 2i$ D $1 + 2 - i$, $1 - 2i$

Q190 MCQ

Find two numbers whose sum is 4 and product is 2 Select the correct answer:

Choose the correct option.

A $2 + \sqrt{2}$, $2 - \sqrt{2}$ (sign error)

B $2 + \sqrt{2}$, $2 - \sqrt{2}$

C $2 + \sqrt{2}$, $2 - \sqrt{3}$

D $2 + \sqrt{2}$, $2 - \sqrt{2} - 1$

Classified Practice

Quadratic Equations

Q191 Written

If α, β are roots of $x^2 + 2x - 4 = 0$, find a quadratic with coefficient of x^2 equal to 1 and roots $\alpha + 2, \beta + 2$.

Q192 MCQ

If α, β are roots of $x^2 + 2x - 4 = 0$, find a quadratic with coefficient of x^2 equal to 1 and roots $\alpha + 2, \beta + 2$ Select the correct answer:

Choose the correct option.

A $x^2 - 2x - 4 = 0 - 1$

B $x^2 - 2x - 4 = 0$ (sign error)

C $x^2 - 2x - 4 = 0$

D $x^2 - 2x - 4 = 1$

Classified Practice

Quadratic Equations

Q193 Written

If α, β are roots of $x^2 + x - 3 = 0$, find a quadratic with roots α^2, β^2 .

Q194 Written

If α, β are roots of $x^2 - 2x + 7 = 0$, find a quadratic with roots α^2, β^2 .

Classified Practice

Quadratic Equations

Q195 MCQ

If α, β are roots of $x^2 + x - 3 = 0$, find a quadratic with roots α^2, β^2 . Select the correct answer:

Choose the correct option.

A $x^2 - 7x + 9 = 1$

B $x^2 - 7x + 9 = 0 - 1$

C $x^2 - 7x + 9 = 0$ (sign error)

D $x^2 - 7x + 9 = 0$

Q196 MCQ

If α, β are roots of $x^2 - 2x + 7 = 0$, find a quadratic with roots α^2, β^2 Select the correct answer:

Choose the correct option.

A $x^2 + 10x + 49 = 0 - 1$

B $x^2 + 10x + 49 = 0$ (sign error)

C $x^2 + 10x + 49 = 0$

D $x^2 + 10x + 49 = 1$

Classified Practice

Quadratic Equations

Q197 Written

If α, β are roots of $2x^2 + 3x + 4 = 0$, find an integer-coefficient quadratic with roots $\frac{1}{\alpha}, \frac{1}{\beta}$.

Q198 MCQ

If α, β are roots of $2x^2 + 3x + 4 = 0$, find an integer-coefficient quadratic with roots $\frac{1}{\alpha}, \frac{1}{\beta}$ Select the correct answer:

Choose the correct option.

- A $4x^2 + 3x + 2 = 0$
- B $4x^2 + 3x + 2 = 1$
- C $4x^2 + 3x + 2 = 0 - 1$
- D $4x^2 + 3x + 2 = 0$ (sign error)

Classified Practice

Quadratic Equations

Q199 Written

If α, β are roots of $x^2 + 2x + 3 = 0$, find a quadratic with roots $\alpha + 3\beta$ and $3\alpha + \beta$.

Q200 MCQ

If α, β are roots of $x^2 + 2x + 3 = 0$, find a quadratic with roots $\alpha + 3\beta$ and $3\alpha + \beta$ Select the correct answer:

Choose the correct option.

A $x^2 + 8x + 24 = 1$

B $x^2 + 8x + 24 = 0 - 1$

C $x^2 + 8x + 24 = 0$ (sign error)

D $x^2 + 8x + 24 = 0$

Classified Practice

Quadratic Equations

Q201 Written

If $x^2 + 2mx + m + 2 = 0$ has two distinct positive roots, find the range of m .

Classified Practice

Quadratic Equations

Q202 **Written**

If $x^2 + 2(m - 1)x + 2m^2 - 5m - 3 = 0$ has two distinct positive roots, find the range of m .

Classified Practice

Quadratic Equations

Q203 MCQ

If $x^2 + 2mx + m + 2 = 0$ has two distinct positive roots, find the range of m Select the correct answer:

Choose the correct option.

A $-2 < m < -1$

B $-2 > m > -1$

C $+2 < m < -1$

D $-2 < m < 0$

Q204 MCQ

If $x^2 + 2(m - 1)x + 2m^2 - 5m - 3 = 0$ has two distinct positive roots, find the range of m Select the correct answer:

Choose the correct option.

A $-1 < m < -\frac{1}{2} - 1$

B $-1 < m < -\frac{1}{2}$

C $-(-1 < m < -\frac{1}{2})$

D $-1 < m < -\frac{1}{2} + 1$

Classified Practice

Quadratic Equations

Q205 Written

If $x^2 - 2mx + m + 6 = 0$ has two distinct negative roots, find the range of m .

Classified Practice

Quadratic Equations

Q206 **Written**

If $x^2 + kx + 2k + 5 = 0$ has two distinct negative roots, find the range of k .

Classified Practice

Quadratic Equations

Q207 MCQ

If $x^2 - 2mx + m + 6 = 0$ has two distinct negative roots, find the range of m Select the correct answer:

Choose the correct option.

A $+6 < m < -2$

B $-6 < m < -1$

C $-6 < m < -2$

D $-6 > m > -2$

Q208 MCQ

If $x^2 + kx + 2k + 5 = 0$ has two distinct negative roots, find the range of k Select the correct answer:

Choose the correct option.

A $k > 10$ (sign error)

B $k > 10$

C $k > 11$

D $k > 10 - 1$

Classified Practice

Quadratic Equations

Q209 Written

If $x^2 - (m - 3)x - m - 4 = 0$ has roots of different signs, find the range of m .

Q210 Written

If $x^2 + 4kx - k - 3 = 0$ has roots of different signs, find the range of k .

Classified Practice

Quadratic Equations

Q211 MCQ

If $x^2 - (m - 3)x - m - 4 = 0$ has roots of different signs, find the range of m Select the correct answer:

Choose the correct option.

A $m > -4$

B $m > -3$

C $m > -4 - 1$

D $m > -4$ (sign error)

Q212 MCQ

If $x^2 + 4kx - k - 3 = 0$ has roots of different signs, find the range of k Select the correct answer:

Choose the correct option.

A $k > -2$

B $k > -3 - 1$

C $k > -3$ (sign error)

D $k > -3$

Classified Practice

Quadratic Equations

Q213 Challenge

For $x^2 + ax - a = 0$, determine the nature of the solutions as a varies.

Classified Practice

Quadratic Equations

Q214 **Challenge**

For $x^2 - (k + 2)x + k^2 = 0$, determine the nature of the solutions as k varies.

Classified Practice

Quadratic Equations

Q215 Challenge

For $x^2 - 2(a - 1)x + a^2 - 3a + 4 = 0$, determine the nature of the solutions as a varies.

Classified Practice

Quadratic Equations

Q216 Challenge

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $\alpha^2\beta + \alpha\beta^2$.

Classified Practice

Quadratic Equations

Q217 **Challenge**

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $(\alpha - \beta)^2$.

Classified Practice

Quadratic Equations

Q218 Challenge

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $\alpha^3 + \beta^3$.

Classified Practice

Quadratic Equations

Q219 **Challenge**

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $\frac{\beta^2}{\alpha} + \frac{\alpha^2}{\beta}$.

Classified Practice

Quadratic Equations

Q220 Challenge

The quadratics $x^2 + ax + bc = 0$ and $x^2 + bx + ca = 0$ have a non-zero common root, with $a \neq b$ and $c \neq 0$. Show that the other two roots are roots of $x^2 + cx + ab = 0$.

Classified Practice

Quadratic Equations

Q221 **Challenge**

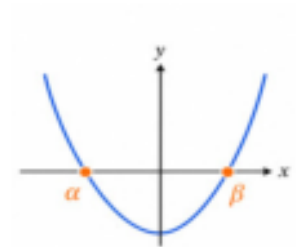
For $S(x) = 6x^4 - 7x^2 - 3$, determine the possible values of x and interpret the real values in the stress model.

Classified Practice

Quadratic Equations

Q222 Challenge

In the opposite figure, α, β are the two roots. If α, β are non-zero roots of a quadratic, construct a quadratic whose roots are $\frac{1}{\alpha}, \frac{1}{\beta}$.



Classified Practice

Quadratic Equations

Q223 **Challenge**

For $x^2 + (m + 1)x + m + 4 = 0$, find the range of m for two distinct real roots that are both positive.

Classified Practice

Quadratic Equations

Q224 **Challenge**

For $x^2 + (m + 1)x + m + 4 = 0$, find the range of m for two distinct real roots that are both negative.

Classified Practice

Quadratic Equations

Q225 **Challenge**

For $x^2 + (m + 1)x + m + 4 = 0$, find the range of m for two roots of opposite signs.

Classified Practice

Quadratic Equations

Q226 Challenge

A model has $R = 3x^2 + 2\sqrt{3}x + 2$. Determine the values of x for which $R = 0$, and interpret the result.

Classified Practice

Quadratic Equations

Q227 Challenge

For $x^2 + ax - a = 0$, determine the nature of the solutions as a varies. Select the correct answer:

Choose the correct option.

- A $a < -4$ or $a > 0$: distinct real; $a = -4, 0$: repeated real; $-4 < a < 0$: distinct real
- B $a < -4$ or $a > 0$: distinct real; $a = -4, 0$: repeated real; $-4 < a < 0$: distinct imaginary
- C $a < -4$ or $a > 0$: repeated real; $a = -4, 0$: repeated real; $-4 < a < 0$: repeated imaginary
- D $a < -4$ or $a > 0$: distinct imaginary; $a = -4, 0$: repeated imaginary; $-4 < a < 0$: distinct imaginary

Classified Practice

Quadratic Equations

Q228 Challenge

For $x^2 - (k + 2)x + k^2 = 0$, determine the nature of the solutions as k varies. Select the correct answer:

Choose the correct option.

- A $-\frac{2}{3} < k < 2$: distinct real; $k = -\frac{2}{3}, 2$: repeated; $k < -\frac{2}{3}$ or $k > 2$: distinct imaginary – 1
- B $-\frac{2}{3} < k < 2$: distinct real; $k = -\frac{2}{3}, 2$: repeated; $k < -\frac{2}{3}$ or $k > 2$: distinct imaginary
- C $-(\frac{2}{3} < k < 2$: distinct real; $k = -\frac{2}{3}, 2$: repeated; $k < -\frac{2}{3}$ or $k > 2$: distinct imaginary)
- D $-\frac{2}{3} < k < 2$: distinct real; $k = -\frac{2}{3}, 2$: repeated; $k < -\frac{2}{3}$ or $k > 2$: distinct imaginary + 1

Classified Practice

Quadratic Equations

Q229 Challenge

For $x^2 - 2(a - 1)x + a^2 - 3a + 4 = 0$, determine the nature of the solutions as a varies. Select the correct answer:

Choose the correct option.

- A $a > 3$: distinct real; $a = 3$: repeated real; $a < 3$: distinct real
- B $a > 3$: distinct real; $a = 3$: repeated real; $a < 3$: distinct imaginary
- C $a > 3$: repeated real; $a = 3$: repeated real; $a < 3$: repeated imaginary
- D $a > 3$: distinct imaginary; $a = 3$: repeated imaginary; $a < 3$: distinct imaginary

Classified Practice

Quadratic Equations

Q230 Challenge

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $\alpha^2\beta + \alpha\beta^2$. Select the correct answer:

Choose the correct option.

A $\frac{9}{2} - 1$

B $\frac{9}{2}$

C $-(\frac{9}{2})$

D $\frac{9}{2} + 1$

Classified Practice

Quadratic Equations

Q231 Challenge

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $(\alpha - \beta)^2$ Select the correct answer:

Choose the correct option.

A 4

B $3 - 1$

C 3 (sign error)

D 3

Classified Practice

Quadratic Equations

Q232 Challenge

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $\alpha^3 + \beta^3$ Select the correct answer:

Choose the correct option.

A $\frac{27}{2} - 1$

B $\frac{27}{2}$

C $-(\frac{27}{2})$

D $\frac{27}{2} + 1$

Classified Practice

Quadratic Equations

Q233 Challenge

If α, β are roots of $2x^2 - 6x + 3 = 0$, find $\frac{\beta^2}{\alpha} + \frac{\alpha^2}{\beta}$ Select the correct answer:

Choose the correct option.

A 9

B 10

C $9 - 1$

D 9 (sign error)

Classified Practice

Quadratic Equations

Q234 Challenge

The quadratics $x^2 + ax + bc = 0$ and $x^2 + bx + ca = 0$ have a non-zero common root, with $a \neq b$ and $c \neq 0$. Show that the other two roots are roots of $x^2 + cx + ab = 0$ Select the correct answer:

Choose the correct option.

- A Common root c; other roots a,b; equation $x^2+cx+ab=0$ (equation = -1).
- B Common root c; other roots a,b; equation $x^2+cx+ab=0$ (sign error).
- C Common root c; other roots a,b; equation $x^2+cx+ab=0$.
- D Common root c; other roots a,b; equation $x^2+cx+ab=0$ (equation = 1).

Classified Practice

Quadratic Equations

Q235 Challenge

For $S(x) = 6x^4 - 7x^2 - 3$, determine the possible values of x and interpret the real values in the stress model. Select the correct answer:

Choose the correct option.

A $x = \pm \frac{\sqrt{6}}{2}$, $x = \pm \frac{i\sqrt{3}}{3}$; real: $\pm \sqrt{6}/2$

B $x = \frac{\sqrt{6}}{2}$, $x = \frac{i\sqrt{3}}{3}$; real: $\sqrt{6}/2$

C $x = -\frac{\sqrt{6}}{2}$, $x = -\frac{i\sqrt{3}}{3}$; real: $-\sqrt{6}/2$

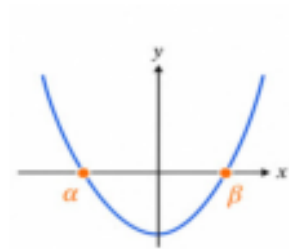
D $x = +\frac{\sqrt{6}}{2}$, $x = +\frac{i\sqrt{3}}{3}$; real: $+\sqrt{6}/2$

Classified Practice

Quadratic Equations

Q236 Challenge

If α, β are non-zero roots of a quadratic, construct a quadratic whose roots are $\frac{1}{\alpha}, \frac{1}{\beta}$. Select the correct answer:



Choose the correct option.

- A If $S = \alpha + \beta$ and $P = \alpha\beta \neq 0$, then $Px^2 - Sx + 1 = 0$.
- B If $S = \alpha + \beta$ and $P = \alpha\beta \neq 0$, then $Px^2 - Sx + 1 = 1$.
- C If $S = \alpha + \beta$ and $P = \alpha\beta \neq 0$, then $Px^2 - Sx + 1 = 0$. -1
- D If $S = \alpha + \beta$ and $P = \alpha\beta \neq 0$, then $Px^2 - Sx + 1 = 0$.; (sign error)

Classified Practice

Quadratic Equations

Q237 Challenge

For $x^2 + (m + 1)x + m + 4 = 0$, find the range of m for two distinct real roots that are both positive. Select the correct answer:

Choose the correct option.

- A $+4 < m < -3$
- B $-4 < m < -2$
- C $-4 < m < -3$
- D $-4 > m > -3$

Classified Practice

Quadratic Equations

Q238 Challenge

For $x^2 + (m + 1)x + m + 4 = 0$, find the range of m for two distinct real roots that are both negative. Select the correct answer:

Choose the correct option.

A $m > 6$

B $m > 5 - 1$

C $m > 5$ (sign error)

D $m > 5$

Classified Practice

Quadratic Equations

Q239 Challenge

For $x^2 + (m + 1)x + m + 4 = 0$, find the range of m for two roots of opposite signs. Select the correct answer:

Choose the correct option.

A $m > -4$

B $m < +4$

C $m < -3$

D $m < -4$

Classified Practice

Quadratic Equations

Q240 Challenge

A model has $R = 3x^2 + 2\sqrt{3}x + 2$. Determine the values of x for which $R = 0$, and interpret the result. Select the correct answer:

Choose the correct option.

- A $x = \frac{-\sqrt{3}i\sqrt{3}}{3}$; no real equilibrium
- B $x = \frac{-\sqrt{3} - i\sqrt{3}}{3}$; no real equilibrium
- C $x = \frac{-\sqrt{3} + i\sqrt{3}}{3}$; no real equilibrium
- D $x = \frac{-\sqrt{3} \pm i\sqrt{3}}{3}$; no real equilibrium

Classified Practice

Quadratic Equations

Quiz MCQ 1 **Concept Quiz · MCQ**

Solve $x^2 - 5x + 6 = 0$ Select the correct answer:

Choose the correct option.

A 2, 3 (sign error)

B 2, 3

C 2, 4

D 2, 3 - 1

Classified Practice

Quadratic Equations

Quiz MCQ 2 **Concept Quiz · MCQ**

For $x^2 - 4x + 7 = 0$, determine the nature of the roots Select the correct answer:

Choose the correct option.

- A Two distinct non-real complex roots
- B Two repeated non-real complex roots
- C Two distinct non-imaginary complex roots
- D Two distinct non-real complex roots;(sign error)

Classified Practice

Quadratic Equations

Quiz MCQ 3 **Concept Quiz · MCQ**

If α, β are roots of $x^2 - 3x + 1 = 0$, find $\alpha^2 + \beta^2$ Select the correct answer:

Choose the correct option.

- A $7 - 1$
- B 7 (sign error)
- C 7
- D 8

Quiz MCQ 4 Concept Quiz · MCQ

Create a monic quadratic with roots $2 + i$ and $2 - i$ Select the correct answer:

Choose the correct option.

A $x^2 - 4x + 5 = 0$

B $x^2 - 4x + 5 = 1$

C $x^2 - 4x + 5 = 0 - 1$

D $x^2 - 4x + 5 = 0$ (sign error)

Classified Practice

Quadratic Equations

Quiz WRITTEN 5 Concept Quiz · Written**Solve** $2x^2 + 3x - 2 = 0$.

Quiz WRITTEN 6 Concept Quiz · Written

Find the range of m for which $x^2 - 2mx + m + 2 = 0$ has two distinct imaginary roots.

Classified Practice

Quadratic Equations

Quiz WRITTEN 7 Concept Quiz · Written

If α, β are roots of $x^2 - 4x + 3 = 0$, find $\alpha^3 + \beta^3$.

Quiz WRITTEN 8 Concept Quiz · Written

If α, β are roots of $x^2 - 2x + 5 = 0$, form the quadratic with roots α^2, β^2 .

Answer key

Use the question number shown on each page.

Q001 $x = \pm 2i\sqrt{2}$

Q002 $x = \pm 2i$

Q003 $x = \pm 7i$

Q004 $x = \pm \frac{2}{3}$

Q005 $x = \pm 2\sqrt{3}$

Q006 $x = \pm \frac{i\sqrt{3}}{2}$

Q007 A

Q008 A

Q009 C

Q010 C

Q011 B

Q012 B

Q013 $x = 3 \pm 2i\sqrt{2}$

Q014 $x = 2 \pm \sqrt{5}$

Q015 $x = -6 \pm i\sqrt{6}$

Q016 $x = -4 \pm 3i\sqrt{2}$

Q017 $x = \frac{3 \pm \sqrt{7}}{2}$

Q018 A

Q019 B

Q020 C

Q021 C

Q022 D

Q023 $x = -1, \frac{9}{2}$

Q024 $x = \frac{1}{2}, -3$

Q025 $x = 0, -\frac{7}{2}$

Q026 $x = \frac{3}{2}, -4$

Q027 $x = 3, 5$

Q028 A

Q029 D

Q030 D

Q031 A

Q032 A

Q033 $x = \frac{2 \pm \sqrt{7}}{3}$

Q034 B

Q035 $x = \frac{-\sqrt{6} \pm 4i}{2}$

Q036 $x = \frac{-\sqrt{3} \pm i\sqrt{7}}{2}$

Q037 $x = \frac{-1 \pm i\sqrt{3}}{2}$

Q038 $x = \frac{5 \pm i\sqrt{11}}{6}$

Q039 $x = \sqrt{3} \pm i\sqrt{2}$

Q040 $x = \frac{-\sqrt{2} \pm i}{3}$

Q041 $x = \frac{1 \pm i\sqrt{7}}{2}$

Q042 $x = \frac{\sqrt{5} \pm i\sqrt{7}}{6}$

Q043 $x = 1 \pm \frac{i\sqrt{6}}{2}$

Q044 B

Q045 C

Q046 D

Q047 B

Q048 D

Q049 B

Q050 C

Q051 C

Q052 B

Q053 Two distinct imaginary (non-real complex) solutions

Q054 $-2 \leq m \leq 6$

Q055 $k < -6$ or $k > 2$: distinct real; $k = -6, 2$: repeated real; $-6 < k < 2$: distinct imaginary

Q056 Two distinct imaginary (non-real complex) solutions

Q057 Two distinct real solutions

Q058 One repeated real solution

Q059 Two distinct real solutions

Q060 One repeated real solution

Q061 Two distinct imaginary (non-real complex) solutions

Q062 One repeated real solution

Q063 Two distinct imaginary (non-real complex) solutions

Q064 Two distinct real solutions

Q065 A

Q066 D

Q067 C

Q068 C

Q069 A

Q070 A

Q071 C

Q072 C

Q073 B

Q074 A

Q075 A

Q076 A

Q077 $-1 < m < 2$

Q078 $a < 2$ or $a > 6$: distinct real; $a = 2, 6$: repeated real; $2 < a < 6$: distinct imaginary

Q079 $a = 3$ or $a = -5$

Q080 $-18 < m < -2$

Q081 $0 \leq m \leq 3$

Q082 C

Q083 C

Q084 A

Q085 D

Q086 C

Q087 $\alpha + \beta = 2, \alpha\beta = -\frac{5}{3}$

Q088 $\alpha + \beta = -\frac{1}{2}, \alpha\beta = -\frac{7}{4}$

Q089 $S = -1, P = 2$

Q090 $S = \frac{7}{3}, P = -2$

Q091 C

Q092 A

Q093 B

Q094 A

Q095 12

Q096 $\frac{81}{2}$

Q097 1

Q098 -1

Q099 -2

Q100 -10

Q101 $\frac{14}{3}$

Q102 -10

Q103 A

Q104 B

Q105 A

Q106 B

Q107 D

Q108 B

Q109 B

Q110 D

Q111 -4

Q112 $\frac{2}{3}$

Q113 $S = 0, P = -\frac{3}{5}$

Q114 $\frac{2}{3}$

Q115 6

Q116 C

Q117 D

Q118 D

Q119 D

Q120 D

Q121 $\frac{5}{2}$

Q122 $\frac{1}{2}$

Q123 11

Q124 $\frac{23}{3}$

Q125 A

Q126 C

Q127 D

Q128 D

Q129 $k = 6, x = 2, 6$

Q130 $m = 8, x = -2, -4$

Q131 $(m, x_1, x_2) = (6, \frac{1}{4}, \frac{1}{2})$ or $(-6, -\frac{1}{4}, -\frac{1}{2})$

Q132 A

Q133 A

Q134 D

Q135 $(m, x_1, x_2) = (2, -2, 3)$ or $(0, -3, 2)$

Q136 $(m, x_1, x_2) = (2, 1, 2)$ or $(-4, -2, -1)$

Q137 $m = 5, x = -5, -1$

Q138 $(m, x_1, x_2) = (-8, \frac{1}{2}, \frac{3}{2})$ or $(8, -\frac{3}{2}, -\frac{1}{2})$

Q139 D

Q140 A

Q141 B

Q142 B

Q143 $m = \frac{24}{25}, x = \frac{4}{5}, \frac{6}{5}$

Q144 A

Q145 $(k, x_1, x_2) = (8, 2, 4)$ or $(-27, -3, 9)$

Q146 B

Q147 $2\left(x - \frac{3+i\sqrt{7}}{4}\right)\left(x - \frac{3-i\sqrt{7}}{4}\right)$

Q148 $(x+1-2i)(x+1+2i)$

Q149 $2\left(x - \frac{3+\sqrt{7}}{2}\right)\left(x - \frac{3-\sqrt{7}}{2}\right)$

Q150 $(x+2-i)(x+2+i)$

Q151 $(x+2-\sqrt{2})(x+2+\sqrt{2})$

Q152 $(x-2-i\sqrt{3})(x-2+i\sqrt{3})$

Q153 $2\left(x - \frac{1+i\sqrt{7}}{4}\right)\left(x - \frac{1-i\sqrt{7}}{4}\right)$

Q154 $3\left(x - \frac{5+i\sqrt{11}}{6}\right)\left(x - \frac{5-i\sqrt{11}}{6}\right)$

Q155 $2\left(x - \frac{1+\sqrt{3}}{2}\right)\left(x - \frac{1-\sqrt{3}}{2}\right)$

Q156 C

Q157 A

Q158 D

Q159 D

Q160 B

Q161 A

Q162 C

Q163 A

Q164 B

Q165 $(x-1)(x+1)(x-i\sqrt{3})(x+i\sqrt{3})$

Q166 $(x-\sqrt{3})(x+\sqrt{3})(x-i\sqrt{2})(x+i\sqrt{2})$

Q167 $(x-\sqrt{3})(x+\sqrt{3})(x-i\sqrt{3})(x+i\sqrt{3})$

Q168 $(x-1)(x+1)(x-\sqrt{2})(x+\sqrt{2})$

Q169 B

Q170 B

Q171 D

Q172 C

Q173 $2x^2 - 2x + 1 = 0$

Q174 $\frac{-1+i\sqrt{7}}{2}, \frac{-1-i\sqrt{7}}{2}$

Q175 $x^2 - 6x + 13 = 0$

Q176 $\frac{1+i\sqrt{3}}{2}, \frac{1-i\sqrt{3}}{2}$

Q177 $x^2 - x - 2 = 0$

Q178 $x^2 - 2x + 17 = 0$

Q179 $3x^2 - 10x + 6 = 0$

Q180 $1+2i, 1-2i$

Q181 $2+\sqrt{2}, 2-\sqrt{2}$

Q182 B

Q183 B

Q184 B

Q185 A

Q186 A

Q187 D

Q188 A

Q189 B

Q190 B

Q191 $x^2 - 2x - 4 = 0$

Q192 C

Q193 $x^2 - 7x + 9 = 0$

Q194 $x^2 + 10x + 49 = 0$

Q195 D

Q196 C

Q197 $4x^2 + 3x + 2 = 0$

Q198 A

Q199 $x^2 + 8x + 24 = 0$

Q200 D

Q201 $-2 < m < -1$

Q202 $-1 < m < -\frac{1}{2}$

Q203 A

Q204 B

Q205 $-6 < m < -2$

Q206 $k > 10$

Q207 C

Q208 B

Q209 $m > -4$

Q210 $k > -3$

Q211 A

Q212 D

Q213 $a < -4$ or $a > 0$: distinct real; $a = -4, 0$: repeated real; $-4 < a < 0$: distinct imaginary

Q214 $-\frac{2}{3} < k < 2$: distinct real; $k = -\frac{2}{3}, 2$: repeated; $k < -\frac{2}{3}$ or $k > 2$: distinct imaginary

Q215 $a > 3$: distinct real; $a = 3$: repeated real; $a < 3$: distinct imaginary

Q216 $\frac{9}{2}$

Q217 3

Q218 $\frac{27}{2}$

Q219 9

Q220 Common root c ; other roots a and b .Q221 $x = \pm \frac{\sqrt{6}}{2}$, $x = \pm \frac{i\sqrt{3}}{3}$; real: $\pm \sqrt{6}/2$ Q222 For $P = \alpha\beta \neq 0$: $Px^2 - Sx + 1 = 0$ Q223 $-4 < m < -3$ Q224 $m > 5$ Q225 $m < -4$ Q226 $x = \frac{-\sqrt{3} \pm i\sqrt{3}}{3}$; no real equilibrium

Q227 B

Q228 B

Q229 B

Q230 B

Q231 D

Q232 B

Q233 A

Q234 C

Q235 A

Q236 A

Q237 C

Q238 D

Q239 D

Q240 D

Quiz MCQ 1 B

Quiz MCQ 2 A

Quiz MCQ 3 C

Quiz MCQ 4 A

Quiz WRITTEN 5 $x = \frac{1}{2}, -2$ Quiz WRITTEN 6 $-1 < m < 2$

Quiz WRITTEN 7 28

Quiz WRITTEN 8 $x^2 + 6x + 25 = 0$